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ResearchGate

Ilaria Fontana

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate, University of Arizona — *08/2024–present*
Department of Mathematics — Tucson, AZ (USA)
- Mentor: Lise-Marie Imbert-Gérard

Visiting Assistant Professor, Northwestern University — *09/2022–08/2024*
Department of Engineering Science and Applied Mathematics (ESAM) — Evanston, IL (USA)

Ph.D. Research Engineer, EDF (main French electric company) — *12/2018–02/2022*
R&D Division Lab Paris-Saclay — Palaiseau (France)
- Industrial supervisor: Kyrylo Kazymyrenko

EDUCATION

Ph.D. degree, University of Montpellier — *12/2018–03/2022*
Mathématiques et Modélisation (Mathematics and Modelisation) — Palaiseau & Montpellier (France)
- Collaboration with EDF (main French electric company) with CIFRE agreement, an industial agreement for training through research
- Thesis title: *Interface problems for dam modeling* (*Problèmes d'interfaces pour les ouvrages hydrauliques*)
- Advisor: Daniele A. Di Pietro (University of Montpellier)
- Industrial supervisor: Kyrylo Kazymyrenko (EDF Lab Paris-Saclay)

Master degree, University of Udine — *10/2016–10/2018*
Applied Mathematics — Udine (Italy)
- Thesis title: *Numerical Bifurcation of Equations with Infinite Delay via Pseudospectral Collocation*
- Supervisor: Rossana Vermiglio
- Final grade: 110/110 cum laude (full marks with honors)

Bachelor degree, University of Udine — *09/2013–10/2016*
Mathematics — Udine (Italy)
- Thesis title: *Il teorema del passo di montagna* (*Mountain Pass Theorem*)
- Supervisor: Rodica Toader
- Final grade: 110/110 cum laude (full marks with honors)

RESEARCH INTERESTS

- Numerical methods for PDEs (FEM, Trefftz and quasi-Trefftz methods, tau spectral methods)
- A posteriori error analysis and adaptive algorithms
- Contact problems
- Solid mechanics and electromagnetism
- Constitutive relations for joints and materials

PUBLICATIONS

Articles and Preprints

- I. Fontana, L.-M. Imbert-Gérard, *Quasi-Trefftz spaces for the first-order time-harmonic Maxwell's equations*, 2026.
  - [arXiv]
- I. Fontana, G. Bacquaert, D. A. Di Pietro, K. Kazymyrenko, *Hyperelastic nature of the Hoek–Brown criterion*, European Journal of Mechanics - A/Solids, 2026, **115**:105782.
  DOI: 10.1016/j.euromechsol.2025.105782
  - [arXiv], [HAL]
- I. Fontana, L.-M. Imbert-Gérard, *Generalized Plane Wave quasi-Trefftz spaces for wave propagation in inhomogeneous media*, Proceedings of the 11th Convention of the European Acoustics Association Forum Acusticum / EuroNoise 2025, 2025. DOI: 10.61782/fa.2025.0967
  - [arXiv]
- I. Fontana, D. A. Di Pietro, *An a posteriori error analysis based on equilibrated stresses for finite element approximations of frictional contact*, Computer Methods in Applied Mechanics and Engineering, 2024, **425**:116950. DOI: 10.1016/j.cma.2024.116950
  - [arXiv], [HAL]
- D. A. Di Pietro, I. Fontana, K. Kazymyrenko, *A posteriori error estimates via equilibrated stress reconstructions for contact problems approximated by Nitsche's method*, Computers & Mathematics with Applications, 2022, **111**:61–80. DOI: 10.1016/j.camwa.2022.02.008
  - [arXiv], [HAL]

Works in progress

- I. Fontana, L.-M. Imbert-Gérard, *A quasi-Trefftz space for the time-harmonic Maxwell's equation*
  - Revised manuscript in preparation. Contribution: numerical implementation and validation
- I. Fontana, L.-M. Imbert-Gérard, *Bessel-based approximation of variable-coefficient Helmholtz solutions*
  - In preparation
- K. Burns, I. Fontana, D. Lecoanet, G. Vasil, *Analysis of tau spectrum for 1D eigenvalue problems*
  - In preparation
- I. Fontana, L.-M. Imbert-Gérard, S. Tordeux, *Existence, uniqueness, regularity result for time-harmonic first-order hyperbolic systems of Friedrichs type*
  - In preparation.

Thesis

- I. Fontana, *Interface Problems for Dam Modeling (Problèmes d'interfaces pour les ouvrages hydrauliques)*, Ph.D. Thesis. HAL theses

CONFERENCES TALKS & SEMINARS

Invited Talks and Seminars

WCCM-ECCOMAS 2026, 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering *July 2026*
Quasi-Trefftz Spaces for Time-Harmonic Electromagnetism Munich (Germany)

SIAM AN26, 2026 SIAM Annual Meeting *July 2026*
Recent Advances in Quasi-Trefftz Methods Cleveland, OH (USA)

Serena Team Seminar, Inria *January 2026*
Quasi-Trefftz Spaces for Partial Differential Equations Paris (France)

NEMESIS Seminar, IMAG, University of Montpellier *January 2026*
Quasi-Trefftz Spaces for Linear Differential Equations Montpellier (France)

ICOSAHOM 2025, 15th International Conference on Spectral and High Order Methods *July 2025*
Quasi-Trefftz Spaces for Vector-Valued Equations Montréal (Canada)

Analysis, Dynamics, and Applications Seminar in the Department of Mathematics at the University of Arizona *November 2024*
Analysis of the Tau Method for Eigenvalue Problems Tucson, AZ (USA)

Modeling and Computation Seminar in the Department of Mathematics at the University of Arizona *September 2024*
A Posteriori Error Analysis for Contact Problems with Application to Dam Modeling Tucson, AZ (USA)

WCCM-PANAC 2024, 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics *July 2024*
Adaptive mesh algorithm for frictional contact problems based on a posteriori error analysis Vancouver (Canada)

Alumni Talk at the Scuola Superiore Universitaria Di Toppo Wassermann *December 2023*
Metodi numerici per problemi meccanici (Numerical methods for mechanical problems) Udine (Italy)
Invited talk by the Alumni Association

Seminar in the Department of Mathematics at Purdue University *December 2022*
Dam modeling: Application of a posteriori error analysis to contact problems West Lafayette, IN (USA)

Seminar in the CDLab of the University of Udine *June 2021*
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online

Contributed Talks

WAVES2026, 17th International Conference on Mathematical and Numerical Aspects of Wave Propagation *June 2026*
Quasi-Trefftz Spaces for Time-Harmonic Maxwell's Equations Montréal (Canada)

Spring 2026 Finite Element Circus at Tufts University *April 2026*
Quasi-Trefftz Spaces for Maxwell's Equations Medford, MA (USA)

ADMOS 2025, 12th International Conference on Adaptive Modeling and Simulation *June 2025*
A Posteriori Error Analysis for Contact Problems with Application to Dam Modeling Barcelona (Spain)

EFEF2025, 22nd European Finite Element Fair *May 2025*
Recent Advances on Quasi-Trefftz Methods Based on Generalized Plane Waves Trieste (Italy)

CSE25, SIAM Conference on Computational Science and Engineering *March 2025*
Quasi-Trefftz Methods with Applications to Maxwell's Equations Fort Worth, TX (USA)

Spring 2024 Finite Element Circus at Brown University *April 2024*
An equilibrated a posteriori error analysis for frictional contact problems Providence, RI (USA)

ICCCM 2023, 7th International Conference on Computational Contact Mechanics *July 2023*
Application of a posteriori error analysis to contact problems Turin (Italy)

Spring 2023 Finite Element Circus at Bridgewater State University *March 2023*
Dam modeling: Application of a posteriori error analysis to contact problems Bridgewater, MA (USA)

EFEF2020, 18th European Finite Element Fair *September 2021*
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Paris (France)

USNCCM16, 16th U.S. National Congress on Computational Mechanics *July 2021*
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online

ADMOS 2021, 10th International Conference on Adaptive Modeling and Simulation *June 2021*
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online

Congrés Français de Mécanique *August 2019*
Lois mécaniques pour les ouvrages hydrauliques (Behavioral laws for hydraulic structures) Brest (France)

Seminar in the CDLab of the University of Udine *October 2018*
Numerical Bifurcation of Equations with Infinite Delay via Pseudospectral Collocation Udine (Italy)

Seminar in the CDLab of the University of Udine *September 2018*
Laguerre-type orthogonal polynomials and transformation of the Chebyshev nodes Udine (Italy)

Seminar in the CDLab of the University of Udine *July 2018*
A model describing vapour and droplets Udine (Italy)

Conferences Posters

MFEM Community Workshop *September 2026*
Recent Advances in Quasi-Trefftz Methods Atlanta, GA (USA)

Journées scientifiques du GdR MaNu *October 2021*
A posteriori error estimates via equilibrated stress reconstructions for contact problems approximated by Nitsche's method Le Croisic (France)

Colloque National MECAMAT 2019 *January 2019*
A model describing vapour and droplets Aussois (France)

TEACHING EXPERIENCES

University of Arizona *2024–present*

- Fall 2026: MATH 355: Analysis of Ordinary Differential Equations
- Fall 2025: MATH/DATA 363: Introduction to Statistical Methods
  - Flipped course and co-teaching with Joseph Watkins
- Fall 2024: MATH 122B: Calculus I

Northwestern University *2023–2024*

- Winter 2023–Winter 2024: MATH 228-2: Multiple Integration and Vector Calculus
  - Taught four terms (six sections): Winter 2023 (2), Spring 2023, Fall 2023, Winter 2024 (2)
  - Course co-organized with Michael Miksis, Petia Vlahovska and Daniel Lecoanet

FUNDINGS & SCHOLARSHIPS

CIFRE agreement for Ph.D. funding *2018–2022*
Industrial Agreements for Training through Research

Thesis Research fellowship *2018*
Three months of Master thesis research at the Department of Mathematics and Statistics of the University of Helsinki (Finland) in collaboration with prof. Mats Gyllenberg and with the Biomathematics research group.

"Scuola Superiore" (School for Advanced Studies) fellowship *2013–2018*
The School for Advanced Studies of the University of Udine is an institution for higher education besides the university. Admitted students follow additional courses and are examined every year on two occasions. The school covers university student fees and provides a residence for five years (Bachelor and Master degrees).

OUTREACH & SERVICE EXPERIENCES

Seminar Co-organizer, University of Arizona *2025–present*
Co-organizer with faculty members of the "Modeling and Computation" weekly seminar in the Department of Mathematics at the University of Arizona.

Math Postdoctoral President, University of Arizona *2025–present*
Organizer of the postdoc professional development seminar and other postdoc activities in the Department of Mathematics at the University of Arizona.

Minisymposium Co-organizer, CSE25 *March 2025*
Co-organizer with Lise-Marie Imbert-Gérard of the minisymposium "Advances in Discontinuous Galerkin and Trefftz Methods" at the SIAM Conference on Computational Science and Engineering.

WISE Mentor, University of Arizona *2024–2025*
Mentor for a Data Science master's student in the WISE mentorship program of the University of Arizona.

Math Postdoctoral Vice-President, University of Arizona *2024–2025*
Vice-President in the governance of postdocs in the Department of Mathematics of the University of Arizona.

Student tutor, University of Udine *2017–2018*
Student tutor for Bachelor and Master degrees in Mathematics at the University of Udine. In charge of teaching assistance and outreach activities.

Student representative, University of Udine *2014–2018*
Student representative for Bachelor and Master degrees in Mathematics of the University of Udine.

TECHNICAL SKILLS

Programming languages *Python, R, C/C++, FreeFem++, Dedalus, Mathematica, MATLAB, Fortran, SQL*

Other LaTeX, *Beamer, code_aster, Linux*

LANGUAGES

Italian *Mother tongue*

English *Workplace - Daily*

French *Workplace - Daily*

Spanish *High-school*

Portuguese *Basic*